MILITARY SPECIFICATION

ALIGNMENT EQUIPMENT, WHEEL, FOR MILITARY WHEELED VEHICLES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers wheel alignment equipment used to measure wheel alignment, provide for the adjustment and correction of wheel alignment angles, determine tracking (four or more wheels), and provide a method for steering gear centering. The equipment consists of a transportable vehicle rack and all necessary mechanical measuring equipment to do wheel alignment inspection and aid in adjustment of military wheeled vehicles in a field environment.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS), and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-C-843	-	Cushioning Material, Cellulosic.
PPP-T-60	-	Tape, Packaging, Waterproof.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, ATTN: SMCAR-EST-S (R), Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4910

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MIL-C-104	-	Crates, Wood: Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	-	Preservation, Methods of.
MIL-B-121	-	Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-C-3774	-	Crates, Wood; Open 12,000-and 16,000-Pound Capacity.
MIL-P-19834	-	Plates, Identification or Instruction, Metal Foil, Adhesive Backed, General Specification for.
MIL-L-46010	-	Lubricant, Solid Film: Heat Cured, Corrosion Inhibiting.

STANDARDS

FEDERAL

FED-STD-H28	-	Screw-Thread Standards For Federal Services.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of US Military Property.
MIL-STD-889	-	Dissimilar Metals.
MIL-STD-1186	-	Cushioning, Anchoring, Bracing, Blocking and Waterproofing; With Appropriate Test Methods.

2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

US DEPARTMENT OF LABOR

OSHA 2206	-	General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Copies of specifications, standards, and documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 360.02	-	Manual for Machine Tool Gearing.
AGMA 390.03	-	Gear Handbook, Volume 1, Gear Classification, Materials, and Measuring Methods for Unassembled Gears.

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(Application for copies should be addressed to the American Gear Manufacturers' Association, 1901 N. Fort Myer Dr., Arlington, VA 22209.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B30.1 - Jacks.

(Application for copies should be addressed to the American National Standard Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82 - Standard Practice for Commercial Packaging.
ASTM E380-84 - Standard for Metric Practice.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 54 - Cylindrical Gears for General Engineering and for Heavy Engineering - Modules and Diametral Pitches.
ISO 1122/1 - Glossary of Gear Terms - Part 1: Geometrical Definitions.

(Application for copies of ISO documents should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2.1 and 6.2.1).

3.2 Design. The wheel alignment service unit shall be new and shall meet the requirements specified herein. The design shall incorporate a levelable, transportable vehicle rack, and all equipment that is necessary to measure wheel alignment angles including caster, camber, steering axis inclination, toe, turning radius, determine four or more wheel tracking, and centering of the steering gear. No external sources of power shall be required for the operation of any of the unit's equipment, including mechanical, pneumatic, hydraulic, electrical (including the battery of the vehicle being serviced or any other battery), or any external power source. The design of the unit shall allow

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personnel to make the required measurements and adjustments on the vehicle as specified in 3.5. The complete unit, including the vehicle rack, shall be transportable.

3.2.1 Maintainability. The design and assembly of the wheel alignment service unit and the placement of ramp width, leveling, and other adjusting features shall assure that the maintenance of operational capability is reduced to the simplest form, requiring a minimum of skills and common tools. Instructions for disassembly, repair, assembly, and adjustment shall be clear, concise, and definitive in application. Repair parts shall be clearly identifiable and labeled in the instruction manual (see 6.3.1). All parts subject to wear, breakage, or distortion shall be accessible for adjustment, repair, or replacement.

3.2.2 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the producing industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.3 Construction.

3.3.1 Castings and forgings. Castings shall be free from visible blow holes, porosity, hard spots, cracks, and other surface defects. Forgings shall be free from visible scale, cold shuts, mismatchings, and other surface defects. Strength and other essential properties of the castings and forgings shall meet the performance requirements specified herein.

3.3.2 Welding, brazing, and soldering. Welding, brazing, and soldering shall be of a quality which shall sustain all requirements of the welded, brazed, or soldered parts. These operations shall not be employed as repair measures for defective parts.

3.3.3 Surfaces. All castings, forgings, molded parts, stampings, and welded parts shall be clean and free of sand, dirt, fins, sprues, scale, flux, and other harmful or extraneous materials. External working surfaces shall be smooth and free of burrs, with edges rounded or beveled.

3.3.4 Fastening devices. Screws, pins, bolts, and similar parts holding structural members shall be installed with means for adjustment and for preventing loss of tightness. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise deformed to the extent that the parts cannot be readily disassembled and reassembled. Removable pins shall be secured with chain or cable to prevent loss. All fasteners shall be plated for corrosion resistance or be manufactured from inherently corrosion-resistant material.

3.3.5 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standards section referenced therein.

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3.3.6 Gears. All gears and pinion designs selected for use in the unit or its components shall be manufactured to either the English (USA) standard inch diametral pitch per AGMA 360.02 and AGMA 390.03 or Metric (SI) module ratio per the International Organization for Standardization (ISO) ISO 54 and ISO 1122/1 and ASTM E380.

3.3.7 Gage dials. All indicating dials shall be marked in the English system. The toe shall be marked in fractions of an inch and the caster, camber, steering axis inclination, and turning radius shall be marked in degrees (°). The markings shall be permanent and shall be in plain, bold faced characters. All dials and dial markings shall be capable of being easily read.

3.3.8 Lubrication. All bearings (except sealed, permanently lubricated type), mating gears, and other moving parts shall be provided means to assure adequate lubrication. Recirculating systems shall include filters which are cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid levels. Oil holes, grease fittings, and filler caps shall be located to be accessible. The manufacturer shall include and install all lubricants, fluids, greases, and other compounds in accordance with the best commercial practice of the producing industry (see 6.5). The lubrication systems shall be temporarily capped or stopped to prevent loss of lubricants during shipment. The manufacturer shall include a warning notice in the instruction manual and on the equipment to remove the temporary caps or stoppers before start-up or use of the equipment.

3.3.9 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit. When practical, all parts shall be permanently and legibly marked in accordance with MIL-STD-130.

3.3.10 Drainage. All spaces in the rack in which water could collect and remain, and which could not be wiped away, shall be provided with means to drain the water from the space to avoid corrosion.

3.4 Safety and health requirements. Safety devices shall be provided for all parts presenting safety hazards to the operator or bystanders. Except as otherwise specified (see 6.2.1), the wheel alignment service unit shall comply with the standards promulgated under OSHA 2206 applicable to the alignment service unit itself. Exceptions and additional requirements for safety and health shall be as specified (see 6.2.1 and 6.4). The operator's manual shall clearly outline all safety precautions relevant to the assembly, operation, disassembly and repair of the unit.

3.4.1 Warning plates. Warning plates shall be permanently fastened in accordance with either of the following materials and methods:

(a) Adhesive backed aluminum foil plate in accordance with MIL-P-19834.
(b) Adhesive backed plastic plate, or plastic coated plate, having physical characteristics equivalent to the aluminum foil plate specified above. The plastic, or plastic coated plates, shall be capable of meeting the tests specified in MIL-P-19834.

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Warning plates shall be located and marked in the following manner:

(c) Attached to the left side of the vehicle rack near the turning radius gage:

CAUTION
MAXIMUM CAPACITY - 10,000 POUNDS

(d) Attached on the left side of the vehicle rack, near a spirit level indicator:

NOTICE
THIS RACK MUST BE LEVEL WHILE
HOLDING A VEHICLE IN ORDER TO
PERFORM PROPER WHEEL ALIGNMENT

3.5 Performance and product characteristics. The wheel alignment service unit shall incorporate the facilities and capabilities necessary to indicate alignment position of caster, camber, steering axis inclination, toe, turning radius, steering gear centering, and four or more wheel tracking.

3.5.1 Vehicle rack. The vehicle rack shall accommodate the alignment measurements and adjustments specified in 3.5 by providing a flat, level surface for the vehicle to rest upon, and a level plane for use as a measurement reference. The rack shall allow sufficient room for personnel to get under, and around, the vehicle to make the required measurements and adjustments.

3.5.1.1 Rack levelability. The runway surface of the vehicle rack shall be capable of being leveled to within 1/8 inch of perfect level flatness, along its length and width, with or without a vehicle on it. The height leveling capability of the rack shall be at least six inches front to rear and side to side. Hand activated hydraulic devices (such as the hydraulic jacks specified in 3.5.2.1.9) or mechanical devices (e.g., hand screw jacks) that do not employ any other external means of power may be used in the leveling system. Once the rack is leveled, as indicated by the rack mounted spirit levels, the supporting legs shall be capable of being locked at the proper height to maintain that levelness.

3.5.1.2 Capacities. The wheel alignment service unit shall be capable of performing the functions specified herein on vehicles with gross vehicle weights (GVW) up to 10,000 lbs, wheelbase lengths up to 14 feet, and tread widths ranging from 48 to 84 inches without visible permanent deformation, sagging, bending, or damage. Tread width shall be measured from the center of ground contact footprint on one tire tread to the same center of ground contact footprint on the opposite tire and wheel.

3.5.1.3 Overload. The vehicle rack as a whole, including the leveling system, runways, and approach ramps, shall be designed to withstand gross vehicle weights of at least 20,000 pounds and 10,000 pounds axle loading on the turning radius gages without catastrophic failure.

3.5.2 Details of components. The unit shall consist of, but shall not be limited to, the principal components specified in 3.5.2.1 through 3.5.2.5.

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3.5.2.1 Vehicle rack. The vehicle rack shall be capable of being disassembled and hand loaded aboard a truck by two personnel, having no previous instruction other than the manufacturer's instruction manual for guidance, in a total time period not to exceed two hours. After being transported to a new location, the rack shall be capable of being unloaded by hand, reassembled, and releveled by the same two individuals in a total time period not to exceed 2 1/2 hours. With adjustments at the lowest level, the runway surface shall be level in accordance with 3.5.1.1, and shall be no less than 26 inches and no more than 30 inches above the base bearing plates when the rack is resting on a level surface. The rack shall accommodate the capacities listed in 3.5.1.2 without degredation of levelability.

3.5.2.1.1 Ramps. Ramps shall be provided to allow the vehicle to be driven onto the vehicle rack. The ramps shall be able to accommodate, at a minimum, the capacities listed in 3.5.1.2 and shall have an overload capacity in accordance with 3.5.1.3. The travel surface of each ramp shall provide traction for the vehicle but shall be fabricated in such a manner that it shall not clog with mud, clay, or dirt. This may be accomplished by, but is not limited to, the use of channel iron or bar stock. The slope of the ramps shall be adjustable so that the ramps are capable of meeting the runway surfaces at any of the adjustable height settings of the supporting legs. The slope shall not be steeper than one inch of rise for four inches of run when the leveling legs are at maximum extension. The width of the ramp travel surface shall be the same width as the adjustable runways. Guide rails shall be provided as specified in 3.5.2.1.4. The ramps shall be provided with means for being securely fixed to the rack, and with the capability of being removable.

3.5.2.1.2 Runways. Each runway shall have a width of not less than 14 inches, and a length sufficient to handle vehicle wheel base lengths up to 14 feet. Guide rails shall be provided along the inside of each runway as specified in 3.5.2.1.4. Each runway shall be provided with a portable wheel chock and a positively fixed front stop. The runways shall support, without deformation or degredation of levelability, the capacities listed in 3.5.1.2 and the overload conditions specified in 3.5.1.3 without failure or permanent deformation of parts. The design shall allow for easy adjustment for tread width changes without the need for disassembly of the rack or without the use of outside force.

3.5.2.1.3 Turning radius gage base. A "pan" or indentation shall be incorporated into the front of each runway to allow for mounting the turning radius gages. The indentation shall be deep enough so that the surface of the turning radius gage on which the wheel sits is flush with the surface of the runway. The base that the turning radius gage rests upon shall be wider than the runway and shall extend to the outside of the rack a sufficient amount to allow the operator to stand safely upon it when making wheel alignment measurements and adjustments. Other methods to provide the same standing room are acceptable.

3.5.2.1.4 Guide rails. The guide rails shall provide a means of positive indication that the vehicle is at the side of the ramp and in danger of being driven off the rack. The guide rails shall be permanently joined to the inside of each runway and ramp and shall be of sufficient height to prevent the vehicle from accidently being driven off the ramp without interfering with the measurements and adjustments specified in 3.5.

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3.5.2.1.5 Runway stops. The runway stops shall be positioned at the front end of each runway and shall provide positive resistance to wheels to prevent vehicles from rolling forward off the runways. The stops shall be securely fixed to the front end of each runway, and shall not interfere with the measurements and adjustments specified in 3.5.

3.5.2.1.6 Supporting legs. There shall be a minimum of four supporting legs on the rack, one located at each corner of the rack. The supporting legs shall each be adjustable in height six inches to provide the levelability specified in 3.5.1.1. A bearing plate shall be permanently fixed to the bottom of each supporting leg (see 3.5.2.1.7).

3.5.2.1.7 Bearing plates. The base bearing plates shall be of sufficient size to prevent sinking of the rack by dispersing the combined weight of the unit, and the loads specified in 3.5.1.2 on surfaces of soil bearing strength as low as 1500 pounds per square foot.

3.5.2.1.8 Spirit levels. The spirit levels shall be permanently attached to, and indicate the relative levelness of, the vehicle rack. The levels shall be protected from damage. A minimum of one spirit level shall be provided, equidistant, between each two supporting legs on both sides and both ends of the rack. No spirit levels shall be located in vertical alignment with a supporting leg.

3.5.2.1.9 Hydraulic jacks. Two hydraulic cylinder hand jacks of piston cylinder design, capable of being used both on and off the vehicle rack, shall be provided with each unit. The jacks shall be capable of lifting a front axle loading of 10,000 pounds each, and 20,000 pounds as a pair. The jacks shall comply with ANSI B30.1. Supports shall be provided on the vehicle rack so that both jacks may be used simultaneously.

3.5.2.2 Brake pedal depressor. The brake pedal depressor shall provide a positive action means of locking the vehicle's brake pedal in the depressed (brakes locked) position.

3.5.2.3 Steering wheel holder. The steering wheel holder shall provide a positive action means of locking the vehicle's steering wheel in place when the steering gear and steering wheel are centered.

3.5.2.4 Gages.

3.5.2.4.1 Turning radius gages. Two turning radius gages, capable of supporting 10,000 pounds each, shall be provided with the unit. The gages shall be usable both on and off the rack. A pair of plates of the same height shall be provided so that the vehicle's rear wheels can be elevated to a height equal to the front wheels when the turning radius gages are used off the vehicle rack on a level surface. The turning radius gages shall have a minimum range of 30 degrees left to 0 degree to 30 degrees right, shall be graduated in increments no larger than one degree, and shall be marked numerically in increments no larger than five degrees. The gages shall be accurate to $\pm$1/4 degree and shall be readable to 1/2 degree. The turning radius gages shall be designed with fully floating action, enabling the turning plates to simultaneously rotate and move laterally. The turning radius gages shall be provided with pins to lock the gages in a non-moveable position at 0 degree. Bearings and moving parts shall be protected from

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damage resulting from dirt, sand, mud, water, or other contaminants falling from above. Turning radius gage bearings and bearing surfaces shall be coated with dry film lubricant conforming to MIL-L-46010.

3.5.2.4.2 Caster, camber, and steering axis inclination gages. Two sets of gages for caster, camber, and steering axis inclination shall be provided. The gages shall be of the mechanical variety and shall indicate in degrees by means of spirit level devices. The gages shall be attachable to the wheels by means of wheel clamps as specified in 3.5.2.5. The gages shall be as follows:

(a) Caster. The caster gages shall indicate the angle, in a plane parallel to the vehicle centerline, forward or rearward, that a line through the center of the ball joints/kingpins/struts varies from plumb/vertical. The gages shall have a minimum range of minus/negative nine degrees (forward at the top of the ball joints or king pin) to plus/positive nine degrees (rearward), shall be graduated in increments no larger than 1/2 degree, and shall be marked numerically in increments no larger than one degree. The gages shall be accurate within $\pm$3/32 degree from true and shall be readable to 1/8 degree.

(b) Camber. The camber gages shall indicate the angle of tilt at the top of the wheel inward/negative, or outward/positive from plumb vertical. The gages shall have a minimum range of minus three degrees (top inward toward the vehicle engine) to plus three degrees (top outward), shall be graduated in increments no larger than 1/4 degree, and shall be marked numerically in increments no larger than one degree. The gages shall be accurate within $\pm$3/32 degree from true and shall be readable to 1/8 degree.

(c) Steering axis inclination. The steering axis inclination gages shall indicate the angle of tilt of a line through the center of the ball joints, kingpin, or strut toward the center of the vehicle from plumb vertical. The gages shall have a minimum range of minus nine degrees (outward) to plus nine degrees (inward), shall be graduated in increments no larger than 1/2 degree, and shall be marked numerically in increments no larger than one degree. The gages shall be accurate within $\pm$3/32 degree from true and shall be readable to 1/8 degree. The caster, camber, and steering axis inclination gages may be integrated into one or more gages.

3.5.2.4.3 Toe gages. A mechanical or optical wheel toe gage shall be provided to measure the distance (in inches) the tread width varies from the front to the back of the tires. The gage shall be capable of being used on both front and rear wheels ranging from 10 to 24 inches in diameter, and with tread widths from 48 to 84 inches. The toe gage shall be adjustable so that the toe is measured at hub (center) height. The gages shall have a measuring range of at least 1/2 inch toe-out to 0 to 1/2 inch toe-in, be graduated in increments no larger than 1/16 inch, and be marked numerically in increments no larger than 1/4 inch. The gages shall be accurate within $\pm$1/32 inch of true and readable to 1/16 inch. The toe gage shall be usable in conjunction with the turning radius gages. If necessary, a means of scribing a center line around the circumference of the tire shall be provided.

3.5.2.4.4 Four wheel tracking gages. The four wheel tracking gages shall indicate the relative squareness (wheel/axle centers forming a rectangle with 90°

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corners) with respect to each other. The gages shall be mechanical or optical, or a combination thereof. The four wheel tracking condition shall be indicated in inches and shall measure misalignments up to at least one inch on wheel base lengths up to, and including, 252 inches (21 feet). The gages shall be readable and accurate to ±1/16 inch. The four wheel tracking and toe gages may be integrated into one measuring system.

3.5.2.4.5 Gage finish. All gages, except the turning radius gages, shall be inherently corrosion resistant or shall be finished (plated) to resist corrosion.

3.5.2.5 Wheel clamps. One or more wheel clamps, as required for each gage, shall be provided as a means of attaching the gages to the wheels of vehicles. The wheel clamps shall be attachable to the wheels by positive mechanical action. Magnetic type clamps, or any other non-mechanical action clamps, may be supplied in addition to the wheel alignment service unit. Clamps and gages shall not be required to be held or assisted by the operator to remain in place while being used. The clamps shall be able to accommodate wheel diameters ranging from 10 inches to 24 inches. The wheel clamps shall adjust to eliminate runout to establish a true and constant plane at 90 degrees to the wheel spindle/hub bearing.

3.6 Manuals. Two sets of manuals shall be provided with each unit. Each set of manuals shall consist of one manual explaining the theory of wheel alignment based on zero camber with the caster spread principle of accommodating road crown, and one operator's manual with instructions on how to use the supplied equipment in accordance with the wheel alignment theory manual. Manuals shall be forwarded to the Principal Contracting Officer for Government evaluation and approval before mass printing in accordance with the DD1423 forms referenced in the contract (see 6.3 and 6.3.1).

3.7 Painting. Unless otherwise specified (see 6.2.1), painting and finishing shall be in accordance with the best commercial practice in the industry. Before painting, all surfaces shall be clean and free of all foreign matter detrimental to painting. At least one coat of primer (or a phosphate or chromate base) and one coat of enamel, or equivalent (i.e., epoxy enamel), shall be applied.

3.8 Identification of product. The wheel alignment service unit shall be marked for identification in accordance with MIL-STD-130 and shall include the National Stock Number. The vehicle rack requirements for levelness and capacity, as specified in 3.5.1.1 and 3.5.1.2, shall be included in the product markings.

3.9 Workmanship. Standards of workmanship shall assure that the alignment unit shall have the stability, strength, safety, and efficient operating characteristics found in the best commercial units and as specified in Section 3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified

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herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. When specified (see 6.2.1), first article inspection shall be applied to the sample noted in 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.3 and the tests of 4.4.1 and 4.4.2 through 4.4.2.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be applied to each production unit offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of (a) through (e) below. Failure of any unit to pass an examination or test shall be cause for rejection of the unit.

(a) Product examination (see 4.3).
(b) Gage accuracy test (see 4.4.1).
(c) Leveling accuracy test (see 4.4.2.2).
(d) Load test (see 4.4.2.4).
(e) Packaging inspection (see 4.5).

4.2.2.1 Quality conformance sampling. Sampling for quality conformance inspection shall be in accordance with MIL-STD-105, using the following Inspection Levels and Acceptable Quality Levels (AQL):

Inspection of Test	Inspection Level	AQL
Product examination	II	1.0
Gage accuracy test	II	1.0
Leveling accuracy test	II	1.0
Load test	II	1.0
Packaging inspection	S-4	4.0

4.3 Product Examination. Visually and manually examine the wheel alignment unit to determine conformance with the requirements of 3.2 through 3.4.1 and

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3.5.2.1 through 3.9. Visual examination shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from defects. Manual examination shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the equipment has passed the examinations prescribed for it by the contractor's inspection system shall be cause for rejection. A certificate of conformance will be accepted in lieu of inspection for the antistatic compound requirement of 3.5.1.

4.4 Test methods.

4.4.1 Gage accuracy test. Each gage and measuring device furnished with the wheel alignment service unit shall be tested for accuracy by comparison with standard measuring equipment which is accurate to within 25 percent, or better, of the accuracy limits specified for the gage or measuring device (see 3.5.2.4.1 through 3.5.2.4.4).

4.4.2 Rack functional test. For first article inspection only (see 4.2.1), the following tests shall be performed on unpaved soil which is flat and level within six inches and has a bearing strength of no more than 1500 pounds per square foot (see 3.5.2.1.7). A report from a soils laboratory certifying the soil bearing strength shall be provided the government.

4.4.2.1 Set up and leveling test. The rack shall be transported to the test site by truck, unloaded, set up, and leveled by two persons unfamiliar with the rack, using the instructions in the manuals provided (see 3.6). Unsuitability of the rack for unloading, set up, and leveling by two persons in accordance with the instructions provided in a total time of two and one half hours or less shall be cause for rejection (see 3.5.2.1).

4.4.2.2 Leveling accuracy test. After the rack has been set up and leveled, the accuracy of the leveling system on the rack shall be verified by measuring the levelness of each runway surface and the levelness of the runway surfaces with respect to each other. Verification shall be made using an inclination measuring device which is accurate to within 30 seconds of a degree. Failure of the leveling system on the rack to indicate levelness of the runway surfaces in accordance with 3.5.1.1 shall be cause for rejection.

4.4.2.3 Vehicle alignment test. A vehicle with known steering misalignments shall be driven onto the rack. With the vehicle on the rack, secondary leveling of the rack shall be performed as necessary using the instructions provided by the manufacturer. While on the rack, the vehicle shall be checked for misalignment and degree of correction necessary for caster, camber, steering axis inclination, toe, turning radius, steering gear centering, and wheel tracking. Necessary adjustments shall be made to correct caster, camber, and toe. Failure of the alignment unit gages to correctly indicate misalignment of caster, camber, steering axis inclination, toe, turning radius, steering gear centering, or wheel tracking without external power; or failure of the alignment unit to provide sufficient working room so that necessary adjustments can be made shall be cause for rejection (see 3.2, 3.5, 3.5.1, 3.5.2.4.1 through 3.5.2.4.4, and 3.5.2.5).

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4.4.2.4 Load test. A vehicle of at least 10,000 pounds GVW shall be rolled or driven onto the rack, parked for five minutes, and rolled or driven back off the rack. Any visible permanent deformation of, or structural damage to, any part of the rack, including the ramps and runways, shall be cause for rejection (see 3.5.1.2).

4.4.2.5 Overload test. A vehicle of at least 20,000 pounds GVW shall be rolled or driven onto the rack, parked for five minutes, and rolled or driven back off the rack. Distribution of the vehicle load shall be such that the total load imposed on the turning radius gages is at least 10,000 pounds. Fracture or sudden collapse of any part of the rack, including the ramps and runways, or deformation so severe that the vehicle cannot be rolled or driven onto and off the rack by means of the ramps and runways shall be cause for rejection (see 3.5.1.3).

4.4.2.6 Disassembly test. The rack shall be disassembled and loaded onto a truck by two persons unfamiliar with the rack using the instructions in the manuals provided (see 3.6). Unsuitability of the rack to permit disassembly and loading by two persons, in accordance with the instructions provided, in a total time of two hours or less shall be cause for rejection (see 3.5.2.1).

4.5 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Disassembly. The wheel alignment service unit may be disassembled to the extent necessary to assure that all surfaces requiring a preservative are accessible for processing, to assure minimum cube for packing, and to permit reassembly without special skills or tools. Disassembly shall be as specified in MIL-STD-1186.

5.2 Preservation. Preservation shall be level A or Industrial. Unless otherwise specified (see 6.2.1), Industrial preservation shall be provided. When military (Level A) preservation is required, it shall be so specified in the contract.

5.2.1 Level A.

5.2.1.1 Cleaning and drying. The wheel alignment service unit shall be cleaned in accordance with process C-1 and dried in accordance with procedure D-1 or D-4 of MIL-P-116.

5.2.1.2 Preservative application. Immediately after cleaning and drying, a coat of type P-2 preservative shall be applied to all critical ferrous metal surfaces of the wheel alignment service unit and accessories. All remaining ferrous metal surfaces subject to corrosion shall be preserved with an application of type P-1 or P-19 preservative. The preservatives specified and the procedures of application shall be in accordance with MIL-P-116.

5.2.1.3 Unit packaging. Each wheel alignment service unit shall be individually unit packed in accordance with method I of MIL-P-116. Parts preserved with P-2 type preservative shall be overwrapped with barrier material conforming to MIL-B-121, grade A. All critical surfaces shall be protected with

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cushioning material conforming to PPP-C-843, type II, class B. The barrier materials shall be secured with tape conforming to PPP-T-60. Small components, gages, and hardware shall be packaged in one or more fiberboard containers conforming to PPP-B-636, class weather-resistant, and secured to the basic unit. When an open crate is used as the shipping container, the packaged components shall be packed in one or more nailed wood boxes conforming to PPP-B-621, class 1, with the box(s) secured to the base crate.

5.2.2 Industrial. Industrial preservation shall be as specified in ASTM D3951-82.

5.3 Packing. Packing shall be level A or Industrial. Unless otherwise specified (see 6.2.1), Industrial packing shall be provided. When military (Level A) packing is required, it shall be so specified in the contract.

5.3.1 Level A. Each wheel alignment service unit shall be packed in one or more crates conforming to MIL-C-104 or MIL-C-3774. Blocking, bracing, and anchoring of the items shall be in accordance with MIL-STD-1186. When MIL-C-3774 crates are used, all unpainted surfaces of the unit shall be covered with a shroud in accordance with MIL-STD-1186.

5.3.2 Industrial. Industrial packing shall be as specified in ASTM D3951-82.

5.4 Marking. Marking shall be Military or Industrial. Unless otherwise specified (see 6.2.1), Industrial marking shall be provided. When military marking is required, it shall be so specified in the contract.

5.4.1 Military. Military marking shall be as specified in MIL-STD-129.

5.4.2 Industrial. Industrial marking shall be as specified in ASTM D3951-82.

6. NOTES

6.1 Intended use. The wheel alignment units are intended for use by military field maintenance organizations to facilitate the measurement and correction of wheel alignment angles and the detection of frame straightness of military wheeled vehicles, both staff and tactical.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article when required (see 3.1).
(c) Exceptions and additional safety/health requirements when required (see 3.4 and 6.4).
(d) Painting and finishing if different (see 3.7).
(e) First article inspection if different (see 4.2.1).
(f) Quality conformance inspection if different (see 4.2.2).
(g) Level of preservation, packing, and marking if different (see 5.2, 5.3, and 5.4).

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6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract (see 3.2.1, 3.4 and 3.6).

6.3.1 Repair part manuals. Repair part identification, in repair part manuals, should include the prime contractor's name, nomenclature, part number, equipment/part manufacturer (OEM - actual part manufacturer), Federal Supply Code for Manufacturers (FSCM), nomenclature, part number, and price of the part immediately after the prime contractor's part identification. High mortality rate items shall be identified as shown in MIL-M-7298.

6.4 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc., as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

6.5 Standard lubricants. The wheel alignment equipment manufacturer should specify and use to the maximum extent possible those lubricants (see 3.3.8) that are presently in the Department of Defense supply system. MIL-HDBK-113, MIL-HDBK-267, and MIL-STD-838 may be used as selection guides.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - AL
Air Force - 99

User activities:

Army - AT
Air Force - 82

Preparing activity:

Army - AL

Project No. 4910-0532

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-14295C	Alignment Equipment, Wheel, for Military Wheeled Vehicles

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

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